INCH-POUND

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21 July 2017
SUPERSEDING
MIL-DTL-32505
13 November 2014

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM, ALLOYS 7017 and 7020, WELDABLE

This specification is approved for use by the Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers two wrought aluminum armor plate alloys for welded and non-welded applications in nominal thicknesses from 0.500 to 4.000 inch, inclusive (see 6.2). The weldability of wrought aluminum alloy AA7017 armor has been determined for Class I armor for these thicknesses, only; 1.000” and 1.500”. Class I (AA7017) material is a direct replacement for MIL-DTL-46063H w/Amendment 2 material, namely AA7039 for new design and if specified (see 6.2), for old design or repair/replacement. The weldability of wrought aluminum alloy AA7020 armor has been determined for Class II armor for these thicknesses, only; 1.000” and 1.500”.

1.2 Weldability. The material covered by this specification has been demonstrated to be weldable to itself and other weldable alloys (see 6.4).

1.3 Classification. The wrought aluminum armor should be of the following classes, as specified (see 6.2).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army Research Laboratory, Weapons and Materials Research Directorate, Specifications and Standards Office, Attn: RDRL-WMM-C, Aberdeen Proving Ground, MD 21005-5069 or emailed to richard.j.squillacioti.civ@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 7017 aluminum alloy. The applicable gauge range for Class I is 0.500 - 4.000 inches.

1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 7020 aluminum alloy. The applicable gauge range for Class II is 0.500 - 4.000 inches.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V_{50} Ballistic Test for Armor

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

THE ALUMINUM ASSOCIATION, INC.

AA H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at http://www.aluminum.org).

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E34	-	Standard Test Methods for Chemical Analysis of Aluminum and Aluminum-Base Alloys

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ASTM E716	-	Standard Practices for Sampling and Sample Preparation of Aluminum and Aluminum Alloys for Determination of Chemical Composition by Spark Atomic Emission Spectrometry
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Spark Atomic Emission Spectrometry
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at http://www.astm.org.)

SAE INTERNATIONAL

AMS 2750	-	Pyrometry

(Copies of these documents are available online at http://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical composition of the plates shall be within the limits shown in Table I. The limits specified in Table I were taken from Teal Sheets from the Aluminum Association dated 2009-03-01 registered composition limits for Aluminum Alloys. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal transverse (LT) direction shall meet the minimum mechanical properties listed in Table II. If mechanical property requirements differ from those contained in Table II, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

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TABLE I. Chemical composition, weight percent. [1/]

ELEMENTS	SYMBOL	Class I 7017 ALLOY [2/]	Class II 7020 ALLOY [2/]
Silicon	Si	0.35	0.35
Iron	Fe	0.45	0.40
Copper	Cu	0.20	0.20
Manganese	Mn	0.05 - 0.50	0.05 – 0.50
Magnesium	Mg	2.0 - 3.0	1.0 - 1.4
Chromium	Cr	0.35	0.10 – 0.35
Nickel	Ni	0.10	N/A
Zinc	Zn	4.0 – 5.2	4.0 - 5.0
Titanium	Ti	0.15	N/A
Zirconium	Zr	0.10 – 0.25	0.08 – 0.20
Other, min	(Mn + Cr)	0.15	N/A
Other	(Zr + Ti)	N/A	0.08 – 0.25
Other, max. Each	---	0.05	0.05
Other, max. Total [3/]	---	0.15	0.15
Aluminum	Al	Remainder	Remainder

[1/] Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
[2/] Where single units are shown, except for [(Mn + Cr)] these indicate the maximum amounts permitted.
[3/] The sum of those "others" metallic elements equaling 0.010 percent or more for each, shall be expressed to the second decimal before determining the sum.

3.4 Weldability. The vehicle fabricator (OEMs and kit manufacturers, etc.) are required to demonstrate weldability (which includes ballistic shock testing) for armor to armor and armor to non-armor applications in accordance with the Ground Combat Vehicle Welding Code (GCVWC) - Aluminum #19207-12472301. Since the alloys referenced by this specification are not specified in the GCVWC 12472301 or in the revised Military Standard for Weldability of Aluminum Alloys (MIL-STD-1946) it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3. Aluminum fabricators shall qualify a weld procedure for any new armor material being used for armor applications and this requires ballistic shock testing. This requirement (ballistic shock testing and the associated striking velocities) are listed in Appendix A and shall also be specified in the contract or drawing for the vehicle, if the alloys specified by this specification are to be welded at the following thicknesses; Class I material in the thicknesses of 1.000" and 1.500" and Class II material in the thicknesses of 1.000" and 1.500".

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix B. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.5.1 Resistance to spalling. Class II (AA7020) material shall be free of spalling for those impact velocities of the V50 ballistic limit. The size of the spall that shall be considered a spall failure shall be no more than two calibers of the impacting projectile unless otherwise specified in the contract or

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purchase order (see 6.2). Spall that arises as part of the V50 ballistic limit shall be noted on the test data sheet, and test plates considered a failure due to spall shall also be reported as a failure to the ballistic requirements of specification MIL-DTL-32505 (see B.4.2).

3.6 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and shall be such as to enable the material to meet the requirements of these specifications.

3.7 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or order (see 6.2).

TABLE II. Minimum mechanical properties. 1/

Thickness, inches	Tensile Strength, ksi		Yield Strength, 0.2% Offset, ksi		Elongation percent	
	Class I	Class II	Class I	Class II	Class I	Class II
0.500 to 1.500, incl.	63.0	52.2	53.0	45.0	9	10
1.501 to 4.000, incl.	60.0	49.3	50.0	42.0	8	8

1/ The gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.

3.7.1 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by AA H35.2, except for thickness.

3.7.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in Table III.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)								
		OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
		THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)								
0.500	1.000		0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.000	1.575		0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362		0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.000		0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.000	4.000		0.090	0.090	0.105	0.110	0.125	---	---	---

3.8 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters

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shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.8.1 Ballistic test plates. In addition to the markings in 3.8, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see B.5.2).

3.9 Ballistic test plate information. For each lot of aluminum alloy armor a properly completed Aluminum Armor Test Data Form (See Figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.10 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use.

3.10.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

3.11 Stress Corrosion Cracking Resistance. Unless otherwise specified in the contract or order (see 6.2) plate material 0.72 inch and thicker in the short transverse direction shall be resistant to stress corrosion cracking. A minimum of 5 of the 9 specimens tested shall show no evidence of cracking at the end of 96 hours when tested as specified in 4.7.4.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate of the same alloy; same source of molten metal representing the ingots poured, and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit.

4.3 First article inspection. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The first article samples shall be produced using the mill production practice proposed for use in production.

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY 7017 / 7020 ARMOR												
FIRING RECORD:								DATE:				
Plate MANUFACTURER / PRODUCER:								PRIME CONTRACTOR:				
Name:								Name:				
Address:								Address:				
POC:								POC:				
Phone No:								Phone No:				
Fax No:								Fax No:				
SPECIFICATION: MIL-DTL-32505								REVISION:		AMENDMENT:		
CONTRACT NO:								ATC PROJECT NO:				
DCAS REGION:								BALLISTIC TEST CONTRACT NO:				
TEST ITEM IDENTIFICATION:												
Lot No. Class:			Plate No.			Ordered Thickness				Alloy and Temper		
PURPOSE: ___ Acceptance ___ First Article ___ Development												
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample_____)												
CHEMICAL ANALYSIS:			OTHERS: Mn+Cr:			Zr+Ti:				Al :		
Si	Fe	Cu	Mn	Mg	Cr	Zn	Ti	Zr	Ni	Other, max.		
										Each	Total	
MECHANICAL PROPERTIES: AA7017/AA7020 Density :												
UTS (ksi):			0.2% YS (ksi):			Elongation (%):						
WELDABILITY: Thickness: Striking Velocity: Required: Actual: Pass / Fail Thickness: Striking Velocity: Required: Actual: Pass / Fail												

BALLISTIC TEST RESULTS			Actual Thickness (in)	Required V_{50} (fps)	Actual V_{50} (fps)	Pass/ Fail	Notes
Test	Projectile	Obl.(deg)					

SPALL:	Size of spall:	Two Caliber of Projectile:	Pass / Fail:

LOTS REPRESENTED BY:	Reduced Testing	Audit Testing

Lot [met] [failed to meet] the ballistic requirements of specification MIL-DTL-32505.

Government Representative	Date	Supplier Representative	Date

FIGURE 1. Aluminum Armor Test Data Form.

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4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix B. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2).

4.5.1.4 Stress Corrosion Cracking. Nine stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.4.

4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with Table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.3.

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TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.4 Stress Corrosion Cracking. Nine test specimens shall be removed from one plate from each lot for testing in accordance with 4.7.4.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.8) and workmanship (see 3.10) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.7 in accordance with the sampling procedures approved by the procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

4.7 Test specimens.

4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E34, E716, and E1251. In case of dispute, analysis by method E34 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal transverse (LT) direction. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Ballistic testing. The ordered thickness specified in the contract shall be used to determine the test projectile in accordance with Table V. Ballistic testing shall be in accordance with Appendix B. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with Table V and Appendix B to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an extrapolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL (P) for that plate.

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4.7.3.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase orders (see 6.2), the ballistic test plates shall be forwarded to the Commander, U.S. Army Aberdeen Test Center, 400 Colleran Road. Bldg. 358, ATTN: CSTE-DTC-AT-SL-V (K. Beavers), Armor Acceptance – B690, Aberdeen Proving Ground, MD 21005-5059 for ballistic testing for first article or lot acceptance.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	TABLE	CLASS I (7017)	CLASS II (7020)
0.500 - 0.749	Cal. .30 AP M2	30	B-I	Listed	Listed
0.750 - 0.950	Cal. .50 FSP	0	B-II	Listed	Listed
0.951 - 1.500	20-mm FSP	0	B-III	Listed	Listed
0.750 - 1.500	Cal. .30 AP M2	0	B-IV	Listed	Listed
1.501 - 2.500	Cal. .50 AP M2	0	B-V	Listed	Listed
2.501 - 3.500	14.5-mm BS41	0	B-VI	Listed	Listed
3.501 – 4.000	14.5-mm BS41	0	B-VI	TBD 1/	Listed

1/ To be determined.

4.7.3.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.7.4 Stress Corrosion. Short transverse stress corrosion test specimens shall be prepared and tested in accordance with procedure outlined in ASTM G38 and ASTM G47. The specimens shall be stressed to 35 ksi. Nine specimens shall be tested per sample per lot. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.8 Rejection and retest. Unless otherwise specified in the contract or purchase order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic. Rejection and retest of ballistic test plates shall be in accordance with B.5.2.

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4.8.3 Stress corrosion. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense submit thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.4. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.3 and 4.7.4.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing that the manufacturer agrees in writing to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components are to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat and tactical vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Class II (AA7020) has better inherit resistance to spall.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3).
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties and ballistic requirements if different (see 3.3).
(e) Size of spall that ise considered a failure for AA7020 (see 3.5.1 and B.4.2).
(f) Dimension and tolerance requirements if different (see 3.7 and 3.7.1).
(g) If markings are different and when impression stamping is permissible (see 3.8).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.10.1)
(i) If stress corrosion cracking resistance is not required or is different (see 3.11).
(j) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(k) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).

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(l) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(m) If approval was requested and received for a different ballistic testing facility (see 4.7.3.1)
(n) Rejection and retest requirement, if different (see 4.8).
(o) If reduced testing is allowed (see 4.9).
(p) Packaging requirements (see 5.1).
(q) Striking velocities for those thicknesses not covered (see A.3.6.1.1).

6.3 Weldability. The Army Research Lab has conducted testing and found that Class I, AA7017 and Class II, AA7020 alloys can be welded and demonstrate good shock resistance when tested with the method described in the Ground Combat Vehicle Welding Code, TACOM Drawing 19207-12472301. However, 19207-12472301 does not currently address these alloys. Appendix A provides the additional details to permit welding qualification for armor applications. The fabricator performing welding on these alloys for armor applications is responsible for developing and qualifying welding procedures in accordance with 19207-12472301. Appendix A contains the supplemental information for the welding code. The thickness of the weldments specified in the contract or drawing will determine the proofing projectile and striking velocity in accordance with Appendix A. Tests have demonstrated that material covered by this specification is weldable to itself and other weldable 7000, 5000, and 6000 series alloys using a 5000 series filler metal. It is not weldable to 2000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated prior to acceptance by the procuring activity.

6.4 Weld wire. Butt joints were successfully welded with AA5556 weld wire for Class I (AA7017) when tested with the required proofing projectile per the Ground Combat Vehicle Welding Code 19207-12472301. For Class II (AA7020) flat weldments (butt joints) AA5087 and AA5556 weld wires were successfully used in welding the plates for proof testing (see 1.2).

6.5 Density. The density of AA7017 is 0.09971 lb. /in^3 (2.76 g/cm^3). The density of AA7020 is 0.10043 lb. /in^3 (2.78 g/cm^3).

6.6 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)

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6.7 Subject term (key word) listing.

Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
M1114 HMMWV
Military vehicles
Stress corrosion
14.5-mm API
20-mm FSP

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

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APPENDIX A

ARMOR PLATE, ALUMINUM, ALLOY 7017 and 7020, WELDABLE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum welding acceptable requirements (striking velocities, etc.) of aluminum alloy armor plate, AA7017 and AA7020 when welded and tested in accordance with the provisions specified in the applicable contract. When the material specified by this specification is to be used in a welded armor application the requirements for weldability should specify the Ground Combat Vehicle Welding Code (GCVWC) - Aluminum 19207-12472301. In lieu of this document (Welding Code) or a welding standard for aluminum armor and non-armor applications this appendix and the information contained herein is intended for information only. Until the Welding Code or a welding standard for aluminum armor and non-armor applications is updated or revised the information contain herein should be referenced for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

TACOM Drawing 19207-12472301	-	Ground Combat Vehicle Welding Code -- Aluminum, Section 10

(Copies of this document are available online at Ground Combat Vehicle Welding Code - Aluminum, dated 6/03/2003 Drawing Number 12472301).

A.3 REQUIREMENTS

A.3.1 Base Material M Number. Alloys 7017 and 7020 shall be classified as M27 materials.

A3.2 Tensile Strength of Welded Aluminum Alloys (Table 5.1 of 19207-12472301). The minimum tensile strength of 7017, Class I welds, shall be 46 ksi. The minimum tensile strength of 7020, Class II welds, shall be 41 ksi.

A3.3 Base Metal – Non Critical Welds. Material conforming to this specification is permitted for non-critical welds (paragraph 8.2 of 19207-12472301).

A3.4 Base Metal - Critical Welds. Material conforming to this specification is permitted for critical welds (paragraph 9.2 of 19207-12472301).

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APPENDIX A

A3.5 Base Metal- Ballistic Welds. Material conforming to this specification is permitted for critical welds (paragraph 10.2 of 19207-12472301).

A.3.6 Ballistic Shock (Weldments) test. Testing shall be in accordance with TACOM Drawing 19207-12472301, Ground Combat Vehicle Welding Code -- Aluminum, Section 10, except that nothing in that drawing shall be construed to supersede or invalidate the requirements of this specification.

A.3.6.1 Striking Velocities.

A.3.6.1.1 Flat Weldments (Butt joints). Striking velocities and proofing projectile shall be chosen from table A-I based on the alloy and thickness of the plate to be impacted. Striking velocities on thicknesses not covered by these tables shall be as agreed upon in the contract or purchase order (see 6.2).

A.3.6.2 Number of Impacts. One fair impact shall be required on the welded butt joint. Should a second round be required, the procedure specified in the TACOM Drawing 19207-12472301 shall be followed.

TABLE A-I. Minimum required striking velocities for Flat Weldments (Butt joints)

ORDERED THICKNESS (Inches)	Plate Proofing Projectile	Class I (AA7017)	Class II (AA7020)
1.000	75-mm Aluminum, M1002A	900 fps	775 fps
1.500	75-mm Aluminum, M1002A	1130 fps	1200 fps

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APPENDIX B

ARMOR PLATE, ALUMINUM, ALLOY 7017 and 7020, WELDABLE

B.1 SCOPE

B.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, AA7017 and AA7020 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS

B.2.1 Government documents.

B.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

B.3 DEFINITIONS

B.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

B.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

B.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

B.3.4 Partial penetration, (PP). Any impact which is not a complete penetration may be considered a partial penetration.

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B.3.5 Witness plate. A thin sheet located behind and parallel to the ballistic test sample which is used to detect penetrating projectiles or spall.

B.3.6 Spalling. The detachment or delamination of a layer of material in the area surrounding the location of impact, which may occur on either the front or rear surfaces of the armor. Spalling may be a threat mechanism even when penetration of the armor itself is not complete.

B.4 REQUIREMENTS

B.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables B-I through B-VI.

B.4.2 Resistance to spalling. Class II (AA7020) material shall be free of spalling for those impact velocities of the V50 ballistic limit. The size of the spall that shall be considered a spall failure shall be no more than two calibers of the impacting projectile unless otherwise specified in the contract or purchase order (see 6.2). Spall that arises as part of the V50 ballistic limit shall be noted on the test data sheet, and test plates considered a failure due to spall shall also be reported as a failure to the ballistic requirements of specification MIL-DTL-32505.

B.5 TESTS

B.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

B.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

B.5.1.2 Protection ballistic limit, BL(P).

B.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

B.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic

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limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

B.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

B.5.2 Rejection and retest of ballistic plates.

B.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

B.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

B.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

B.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

B.5.3 Disposal of ballistic test plates.

B.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

B.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in B.5.3.1.

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APPENDIX B

TABLE B-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.475	1448	1376	0.580	1662	1641	0.685	1865	1791
0.480	1459	1393	0.585	1672	1650	0.690	1874	1797
0.485	1469	1408	0.590	1682	1659	0.695	1883	1802
0.490	1480	1424	0.595	1692	1667	0.700	1893	1808
0.495	1490	1439	0.600	1702	1676	0.705	1902	1813
0.500 [1/]	1500	1453	0.605	1711	1684	0.710	1911	1818
0.505	1511	1468	0.610	1721	1692	0.715	1921	1823
0.510	1521	1481	0.615	1731	1700	0.720	1930	1829
0.515	1531	1495	0.620	1741	1708	0.725	1939	1834
0.520	1541	1508	0.625	1750	1715	0.730	1948	1839
0.525	1552	1521	0.630	1760	1722	0.735	1958	1844
0.530	1562	1533	0.635	1770	1729	0.740	1967	1849
0.535	1572	1545	0.640	1779	1736	0.745	1976	1854
0.540	1582	1557	0.645	1789	1743	0.749 [2/]	1983	1859
0.545	1592	1569	0.650	1798	1749	0.755	1994	1864
0.550	1602	1580	0.655	1808	1756	0.760	2003	1869
0.555	1612	1591	0.660	1817	1762	0.765	2012	1874
0.560	1622	1601	0.665	1827	1768	0.770	2022	1879
0.565	1632	1611	0.670	1836	1774	0.775	2031	1884
0.570	1642	1621	0.675	1846	1780	0.780	2040	1889
0.575	1652	1631	0.680	1855	1786	0.785	2049	1894

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-II. Minimum required ballistic limits – caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.730	1912	1854	0.835	2398	2274	0.940	3007	2740
0.735	1933	1873	0.840	2424	2295	0.945	3040	2763
0.740	1954	1892	0.845	2450	2316	0.950 2/	3073	2787
0.745	1975	1911	0.850	2477	2337	0.955	3106	2810
0.750 1/	1996	1930	0.855	2504	2359	0.960	3140	2834
0.755	2018	1950	0.860	2531	2381	0.965	3174	2858
0.760	2040	1969	0.865	2558	2402	0.970	3208	2882
0.765	2062	1989	0.870	2586	2424	0.975	3243	2906
0.770	2084	2008	0.875	2614	2446	0.980	3278	2930
0.775	2107	2028	0.880	2642	2468	0.985	3313	2954
0.780	2130	2048	0.885	2671	2490	0.990	3348	2978
0.785	2153	2068	0.890	2700	2512	0.995	3384	3003
0.790	2176	2088	0.895	2729	2534	1.000	3420	3027
0.795	2200	2108	0.900	2759	2557	1.005	3446	3052
0.800	2224	2129	0.905	2789	2579	1.010	3482	3077
0.805	2248	2149	0.910	2819	2602	1.015	3519	3102
0.810	2272	2170	0.915	2849	2625	1.020	3556	3127
0.815	2297	2190	0.920	2880	2648	1.025	3593	3152
0.820	2322	2211	0.925	2911	2671	1.030	3630	3177
0.825	2347	2232	0.930	2943	2694	1.035	3668	3202
0.830	2372	2253	0.935	2975	2717	1.040	3706	3227

1/ Specification requirements begin for this ordered thickness.
2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-III. Minimum required ballistic limits – 20mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.930	1284	1319	1.100	1623	1596	1.270	2052	1998
0.935	1293	1326	1.105	1634	1606	1.275	2066	2012
0.940	1302	1333	1.110	1645	1617	1.280	2080	2025
0.945	1311	1340	1.115	1657	1627	1.285	2094	2039
0.951 1/	1321	1347	1.120	1668	1638	1.290	2109	2053
0.955	1329	1354	1.125	1680	1648	1.295	2123	2066
0.960	1338	1361	1.130	1691	1659	1.300	2138	2080
0.965	1347	1368	1.135	1703	1670	1.305	2153	2094
0.970	1356	1376	1.140	1715	1681	1.310	2168	2108
0.975	1366	1383	1.145	1727	1692	1.315	2183	2122
0.980	1375	1390	1.150	1739	1703	1.320	2198	2136
0.985	1385	1397	1.155	1751	1714	1.325	2213	2150
0.990	1394	1404	1.160	1763	1725	1.330	2228	2164
0.995	1404	1412	1.165	1775	1737	1.335	2244	2178
1.000	1414	1419	1.170	1787	1748	1.340	2259	2193
1.005	1424	1427	1.175	1800	1760	1.345	2275	2207
1.010	1433	1435	1.180	1812	1772	1.350	2291	2221
1.015	1443	1442	1.185	1825	1784	1.355	2307	2236
1.020	1453	1451	1.190	1837	1795	1.360	2323	2250
1.025	1463	1459	1.195	1850	1807	1.365	2339	2265
1.030	1473	1467	1.200	1863	1820	1.370	2355	2280
1.035	1484	1475	1.205	1876	1832	1.375	2371	2294
1.040	1494	1484	1.210	1889	1844	1.380	2388	2309
1.045	1504	1493	1.215	1902	1856	1.385	2404	2324
1.050	1515	1501	1.220	1915	1869	1.390	2421	2339
1.055	1525	1510	1.225	1928	1882	1.395	2437	2354
1.060	1536	1519	1.230	1941	1894	1.400	2454	2369
1.065	1546	1529	1.235	1955	1907	1.405	2471	2384
1.070	1557	1538	1.240	1968	1920	1.410	2488	2399
1.075	1568	1547	1.245	1982	1933	1.415	2506	2414
1.080	1579	1557	1.250	1996	1946	1.420	2523	2429
1.085	1590	1567	1.255	2010	1959	1.425	2540	2444
1.090	1601	1576	1.260	2023	1972	1.430	2558	2459
1.095	1612	1586	1.265	2037	1985	1.435	2576	2474

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-III. Minimum required ballistic limits – 20mm fragment simulating projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.440	2593	2490	1.485	2759	2629	1.530	2936	2771
1.445	2611	2505	1.490	2779	2645	1.535	2956	2787
1.450	2629	2521	1.495	2798	2661	1.540	2976	2803
1.455	2648	2536	1.500 [2/]	2817	2676	1.545	2996	2819
1.460	2666	2551	1.505	2837	2692	1.550	3017	2834
1.465	2684	2567	1.510	2856	2708	1.555	3038	2850
1.470	2703	2582	1.515	2876	2723	1.560	3058	2866
1.475	2722	2598	1.520	2896	2739	1.565	3079	2882
1.480	2741	2614	1.525	2916	2755	1.570	3100	2898

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
0.730	1732	1593	0.900	1969	1838	1.070	2189	2089
0.735	1739	1601	0.905	1976	1846	1.075	2196	2096
0.740	1746	1608	0.910	1982	1853	1.080	2202	2103
0.745	1754	1615	0.915	1989	1861	1.085	2208	2109
0.750 1/	1761	1622	0.920	1996	1868	1.090	2214	2116
0.755	1768	1630	0.925	2002	1875	1.095	2221	2123
0.760	1775	1637	0.930	2009	1883	1.100	2227	2130
0.765	1782	1635	0.935	2016	1890	1.105	2233	2136
0.770	1789	1642	0.940	2022	1898	1.110	2239	2143
0.775	1796	1650	0.945	2029	1905	1.115	2245	2149
0.780	1804	1657	0.950	2037	1913	1.120	2252	2156
0.785	1811	1664	0.955	2042	1920	1.125	2258	2162
0.790	1818	1672	0.960	2048	1928	1.130	2264	2169
0.795	1825	1680	0.965	2055	1935	1.135	2270	2175
0.800	1832	1687	0.970	2062	1943	1.140	2276	2182
0.805	1839	1693	0.975	2068	1950	1.145	2282	2188
0.810	1846	1701	0.980	2075	1958	1.150	2288	2194
0.815	1853	1709	0.985	2081	1965	1.155	2294	2200
0.820	1860	1716	0.990	2087	1973	1.160	2301	2207
0.825	1867	1724	0.995	2094	1980	1.165	2307	2213
0.830	1874	1732	1.000	2100	1988	1.170	2313	2219
0.835	1881	1739	1.005	2107	1995	1.175	2319	2225
0.840	1887	1746	1.010	2113	2003	1.180	2325	2231
0.845	1894	1754	1.015	2120	2010	1.185	2331	2237
0.850	1901	1762	1.020	2126	2018	1.190	2337	2243
0.855	1908	1770	1.025	2132	2025	1.195	2343	2249
0.860	1915	1777	1.030	2139	2032	1.200	2349	2255
0.865	1922	1785	1.035	2145	2040	1.205	2355	2261
0.870	1928	1793	1.040	2152	2047	1.210	2361	2267
0.875	1935	1800	1.045	2158	2054	1.215	2367	2273
0.880	1942	1808	1.050	2164	2061	1.220	2373	2278
0.885	1949	1816	1.055	2171	2068	1.225	2379	2284
0.890	1956	1824	1.060	2177	2075	1.230	2385	2290
0.895	1962	1831	1.065	2183	2082	1.235	2391	2296

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.240	2397	2301	1.350	2525	2420	1.460	2649	2529
1.245	2402	2307	1.355	2531	2425	1.465	2655	2534
1.250	2408	2313	1.360	2536	2430	1.470	2660	2539
1.255	2414	2318	1.365	2542	2435	1.475	2666	2543
1.260	2420	2324	1.370	2548	2440	1.480	2671	2548
1.265	2426	2329	1.375	2553	2445	1.485	2677	2553
1.270	2432	2335	1.380	2559	2450	1.490	2682	2558
1.275	2438	2340	1.385	2565	2455	1.495	2688	2562
1.280	2444	2346	1.390	2570	2460	1.500 2/	2693	2567
1.285	2450	2351	1.395	2576	2465	1.505	2699	2572
1.290	2455	2357	1.400	2582	2470	1.510	2704	2577
1.295	2461	2362	1.405	2587	2475	1.515	2710	2581
1.300	2467	2367	1.410	2593	2480	1.520	2715	2586
1.305	2473	2373	1.415	2599	2485	1.525	2721	2590
1.310	2479	2378	1.420	2604	2490	1.530	2726	2595
1.315	2484	2383	1.425	2610	2495	1.535	2732	2600
1.320	2490	2389	1.430	2615	2500	1.540	2737	2605
1.325	2496	2394	1.435	2621	2505	1.545	2743	2610
1.330	2502	2399	1.440	2627	2510	1.550	2749	2614
1.335	2508	2404	1.445	2632	2514	1.555	2755	2619
1.340	2513	2410	1.450	2638	2519	1.560	2762	2624
1.345	2519	2415	1.455	2643	2524	1.565	2768	2628

2/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
1.480	2000	1878	1.660	2139	2012	1.840	2272	2143
1.485	2004	1882	1.665	2142	2016	1.845	2275	2146
1.490	2008	1886	1.670	2146	2019	1.850	2279	2150
1.495	2012	1890	1.675	2150	2023	1.855	2282	2153
1.501 1/	2016	1893	1.680	2154	2027	1.860	2286	2157
1.505	2019	1897	1.685	2158	2030	1.865	2290	2161
1.510	2023	1901	1.690	2161	2034	1.870	2293	2164
1.515	2027	1904	1.695	2165	2038	1.875	2297	2168
1.520	2031	1908	1.700	2169	2041	1.880	2300	2171
1.525	2035	1912	1.705	2172	2045	1.885	2304	2175
1.530	2039	1916	1.710	2176	2049	1.890	2308	2178
1.535	2043	1919	1.715	2180	2052	1.895	2311	2182
1.540	2047	1923	1.720	2184	2056	1.900	2315	2186
1.545	2051	1927	1.725	2187	2059	1.905	2318	2189
1.550	2055	1931	1.730	2191	2063	1.910	2322	2193
1.555	2058	1934	1.735	2195	2067	1.915	2325	2196
1.560	2062	1938	1.740	2198	2070	1.920	2329	2200
1.565	2066	1942	1.745	2202	2074	1.925	2333	2203
1.570	2070	1946	1.750	2206	2078	1.930	2336	2207
1.575	2074	1949	1.755	2210	2081	1.935	2340	2210
1.580	2078	1953	1.760	2213	2085	1.940	2343	2214
1.585	2082	1957	1.765	2217	2089	1.945	2347	2217
1.590	2085	1960	1.770	2221	2092	1.950	2350	2221
1.595	2089	1964	1.775	2224	2096	1.955	2354	2225
1.600	2093	1968	1.780	2228	2099	1.960	2357	2228
1.605	2097	1972	1.785	2232	2103	1.965	2361	2232
1.610	2101	1975	1.790	2235	2107	1.970	2364	2235
1.615	2105	1979	1.795	2239	2110	1.975	2368	2239
1.620	2108	1983	1.800	2243	2114	1.980	2371	2242
1.625	2112	1986	1.805	2246	2117	1.985	2375	2246
1.630	2116	1990	1.810	2250	2121	1.990	2378	2249
1.635	2120	1994	1.815	2253	2125	1.995	2382	2253
1.640	2124	1997	1.820	2257	2128	2.000	2385	2256
1.645	2127	2001	1.825	2261	2132	2.005	2389	2260
1.650	2131	2005	1.830	2264	2135	2.010	2392	2263
1.655	2135	2008	1.835	2268	2139	2.015	2396	2267

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
2.020	2399	2270	2.195	2519	2392	2.370	2635	2511
2.025	2403	2274	2.200	2522	2395	2.375	2638	2514
2.030	2406	2277	2.205	2526	2399	2.380	2641	2517
2.035	2410	2281	2.210	2529	2402	2.385	2644	2521
2.040	2413	2284	2.215	2532	2406	2.390	2648	2524
2.045	2417	2288	2.220	2536	2409	2.395	2651	2527
2.050	2420	2291	2.225	2539	2412	2.400	2654	2531
2.055	2424	2295	2.230	2542	2416	2.405	2657	2534
2.060	2427	2298	2.235	2546	2419	2.410	2661	2537
2.065	2430	2302	2.240	2549	2423	2.415	2664	2541
2.070	2434	2305	2.245	2552	2426	2.420	2667	2544
2.075	2437	2309	2.250	2556	2429	2.425	2670	2547
2.080	2441	2312	2.255	2559	2433	2.430	2674	2551
2.085	2444	2316	2.260	2562	2436	2.435	2677	2554
2.090	2448	2319	2.265	2566	2440	2.440	2680	2557
2.095	2451	2323	2.270	2569	2443	2.445	2683	2561
2.100	2454	2326	2.275	2572	2446	2.450	2686	2564
2.105	2458	2330	2.280	2576	2450	2.455	2690	2567
2.110	2461	2333	2.285	2579	2453	2.460	2693	2571
2.115	2465	2337	2.290	2582	2457	2.465	2696	2574
2.120	2468	2340	2.295	2585	2460	2.470	2699	2577
2.125	2472	2344	2.300	2589	2463	2.475	2702	2581
2.130	2475	2347	2.305	2592	2467	2.480	2706	2584
2.135	2478	2350	2.310	2592	2470	2.485	2709	2587
2.140	2482	2354	2.315	2599	2474	2.490	2712	2591
2.145	2485	2357	2.320	2602	2477	2.495	2715	2594
2.150	2489	2361	2.325	2605	2480	2.500 [2/]	2718	2597
2.155	2492	2364	2.330	2609	2484	2.505	2722	2600
2.160	2495	2368	2.335	2612	2487	2.510	2725	2604
2.165	2499	2371	2.340	2615	2490	2.515	2728	2607
2.170	2502	2375	2.345	2618	2494	2.520	2731	2610
2.175	2505	2378	2.350	2622	2497	2.525	2734	2614
2.180	2509	2382	2.355	2625	2501	2.530	2737	2617
2.185	2512	2385	2.360	2628	2504	2.535	2740	2620
2.190	2516	2388	2.365	2631	2507	2.540	2743	2623

[2/] Specification requirements end for this ordered thickness

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
2.480	2456	2429	2.655	2560	2523	2.830	2662	2617
2.485	2459	2431	2.660	2563	2526	2.835	2664	2620
2.490	2462	2434	2.665	2566	2528	2.840	2667	2622
2.495	2465	2437	2.670	2569	2531	2.845	2670	2625
2.501 1/	2469	2440	2.675	2572	2534	2.850	2673	2628
2.505	2471	2442	2.680	2675	2536	2.855	2676	2631
2.510	2474	2445	2.685	2678	2539	2.860	2679	2633
2.515	2477	2448	2.690	2581	2542	2.865	2682	2636
2.520	2480	2450	2.695	2584	2544	2.870	2684	2639
2.525	2483	2453	2.700	2586	2547	2.875	2687	2641
2.530	2486	2456	2.705	2589	2550	2.880	2690	2644
2.535	2489	2458	2.710	2592	2552	2.885	2693	2647
2.540	2492	2461	2.715	2595	2555	2.890	2696	2649
2.545	2495	2464	2.720	2598	2558	2.895	2699	2652
2.550	2498	2466	2.725	2601	2560	2.900	2701	2655
2.555	2501	2469	2.730	2604	2563	2.905	2704	2657
2.560	2504	2472	2.735	2607	2566	2.910	2707	2660
2.565	2507	2474	2.740	2610	2569	2.915	2710	2663
2.570	2510	2477	2.745	2613	2571	2.920	2713	2666
2.575	2513	2480	2.750	2616	2574	2.925	2716	2668
2.580	2516	2483	2.755	2618	2577	2.930	2718	2671
2.585	2519	2485	2.760	2621	2579	2.935	2721	2674
2.590	2522	2488	2.765	2624	2582	2.940	2724	2676
2.595	2525	2491	2.770	2627	2585	2.945	2727	2679
2.600	2528	2493	2.775	2630	2587	2.950	2730	2682
2.605	2531	2496	2.780	2633	2590	2.955	2732	2684
2.610	2534	2499	2.785	2638	2593	2.960	2735	2687
2.615	2537	2501	2.790	2639	2595	2.965	2738	2690
2.620	2540	2504	2.795	2641	2598	2.970	2741	2692
2.625	2543	2507	2.800	2644	2601	2.975	2744	2695
2.630	2545	2509	2.805	2647	2603	2.980	2746	2698
2.635	2548	2512	2.810	2650	2606	2.985	2749	2700
2.640	2551	2515	2.815	2653	2609	2.990	2752	2703
2.645	2554	2517	2.820	2656	2612	2.995	2755	2706
2.650	2557	2520	2.825	2659	2614	3.000	2758	2709

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
3.005	2760	2711	3.180	2857	2805	3.355	2952	2899
3.010	2763	2714	3.185	2860	2808	3.360	2955	2902
3.015	2766	2717	3.190	2963	2811	3.365	2957	2905
3.020	2769	2719	3.195	2865	2813	3.370	2960	2907
3.025	2772	2722	3.200	2868	2816	3.375	2963	2910
3.030	2774	2725	3.205	2871	2819	3.380	2965	2913
3.035	2777	2727	3.210	2874	2821	3.385	2968	2916
3.040	2780	2730	3.215	2876	2824	3.390	2971	2918
3.045	2783	2733	3.220	2879	2827	3.395	2973	2921
3.050	2786	2735	3.225	2882	2829	3.400	2976	2924
3.055	2788	2738	3.230	2884	2832	3.405	2979	2926
3.060	2791	2741	3.235	2887	2835	3.410	2981	2929
3.065	2794	2743	3.240	2890	2838	3.415	2984	2932
3.070	2797	2746	3.245	2893	2840	3.420	2987	2934
3.075	2799	2749	3.250	2895	2843	3.425	2989	2937
3.080	2802	2752	3.255	2898	2846	3.430	2992	2940
3.085	2805	2754	3.260	2901	2848	3.435	2994	2942
3.090	2808	2757	3.265	2903	2851	3.440	2997	2945
3.095	2810	2760	3.270	2906	2854	3.445	3000	2948
3.100	2813	2762	3.275	2909	2856	3.450	3002	2950
3.105	2816	2765	3.280	2912	2859	3.455	3005	2953
3.110	2819	2768	3.285	2914	2862	3.460	3008	2956
3.115	2822	2770	3.290	2917	2864	3.465	3010	2959
3.120	2824	2773	3.295	2920	2867	3.470	3013	2961
3.125	2827	2776	3.300	2922	2870	3.475	3016	2964
3.130	2830	2778	3.305	2925	2872	3.480	3018	2967
3.135	2833	2781	3.310	2928	2875	3.485	3021	2969
3.140	2835	2784	3.315	2930	2878	3.490	3024	2972
3.145	2838	2786	3.320	2933	2881	3.495	3026	2975
3.150	2841	2789	3.325	2936	2883	3.500 [2/]	3029	2977
3.155	2844	2792	3.330	2938	2886	3.505	3031	2980
3.160	2846	2795	3.335	2941	2889	3.510	3034	2983
3.165	2849	2797	3.340	2944	2891	3.515	3037	2985
3.170	2852	2800	3.345	2947	2894	3.520	3039	2988
3.175	2854	2803	3.350	2949	2897	3.525	3042	2991

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX B

TABLE B-VI. Minimum required ballistic limits - 14.5mm BS41 projectiles at 0° obliquity (continued).

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II		Class I	Class II
3.501 [1/]	TBD	2977	3.680	TBD	3074	3.860	TBD	3171
3.505	TBD	2980	3.685	TBD	3077	3.865	TBD	3174
3.510	TBD	2983	3.690	TBD	3079	3.870	TBD	3176
3.515	TBD	2985	3.695	TBD	3082	3.875	TBD	3179
3.520	TBD	2988	3.700	TBD	3085	3.880	TBD	3182
3.525	TBD	2991	3.705	TBD	3088	3.885	TBD	3184
3.530	TBD	2993	3.710	TBD	3090	3.890	TBD	3187
3.535	TBD	2996	3.715	TBD	3093	3.895	TBD	3190
3.540	TBD	2999	3.720	TBD	3096	3.900	TBD	3192
3.545	TBD	3002	3.725	TBD	3098	3.905	TBD	3195
3.550	TBD	3004	3.730	TBD	3101	3.910	TBD	3198
3.555	TBD	3007	3.735	TBD	3104	3.915	TBD	3200
3.560	TBD	3010	3.740	TBD	3106	3.920	TBD	3203
3.565	TBD	3012	3.745	TBD	3109	3.925	TBD	3206
3.570	TBD	3015	3.750	TBD	3112	3.930	TBD	3209
3.575	TBD	3018	3.755	TBD	3114	3.935	TBD	3211
3.580	TBD	3020	3.760	TBD	3117	3.940	TBD	3214
3.585	TBD	3023	3.765	TBD	3120	3.945	TBD	3217
3.590	TBD	3026	3.770	TBD	3123	3.950	TBD	3219
3.595	TBD	3028	3.775	TBD	3125	3.955	TBD	3222
3.600	TBD	3031	3.780	TBD	3128	3.960	TBD	3225
3.605	TBD	3034	3.785	TBD	3131	3.965	TBD	3227
3.610	TBD	3036	3.790	TBD	3133	3.970	TBD	3230
3.615	TBD	3039	3.795	TBD	3136	3.975	TBD	3233
3.620	TBD	3042	3.800	TBD	3139	3.980	TBD	3235
3.625	TBD	3045	3.805	TBD	3141	3.985	TBD	3238
3.630	TBD	3047	3.810	TBD	3144	3.990	TBD	3241
3.635	TBD	3050	3.815	TBD	3147	3.995	TBD	3243
3.640	TBD	3053	3.820	TBD	3149	4.000 [2/]	TBD	3246
3.645	TBD	3055	3.825	TBD	3152	4.005	TBD	3249
3.650	TBD	3058	3.830	TBD	3155	4.010	TBD	3252
3.655	TBD	3061	3.835	TBD	3157	4.015	TBD	3254
3.660	TBD	3063	3.840	TBD	3160	4.020	TBD	3257
3.665	TBD	3066	3.845	TBD	3163	4.025	TBD	3260
3.670	TBD	3069	3.850	TBD	3166	4.030	TBD	3263
3.675	TBD	3071	3.855	TBD	3168	4.035	TBD	3266

[1/] Specification requirements begin for this ordered thickness.
[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for extrapolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – AS
Air Force - 11

Preparing activity:
Army – MR
(Project 9535-2017-004)

Review activities:
Army – AT, TE
Navy – SH
Air Force – 84, 99
DLA – CQ

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